

Pairwise Comparisons and Effect Sizes

Welch, Hedges g , Benjamini–Hochberg, Mann–Whitney and Cliff δ

343 runs / 9 methods / 4 scenarios

Step-by-step statistical analysis

of social navigation in an agricultural digital twin

Purpose. To update post-hoc analysis using the M4 anchor against eight alternatives, Welch tests with Hedges g and Benjamini–Hochberg correction, while retaining rank effects only where individual data exist.

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1 Statistical method

For comparator C , $g > 0$ means that C is numerically greater than M4:

$$t_W = \frac{\bar{x}_C - \bar{x}_{M4}}{\sqrt{s_C^2/n_C + s_{M4}^2/n_{M4}}}, \quad (1)$$

with Welch–Satterthwaite approximate degrees of freedom. The bias-corrected effect size is

$$g = \frac{\mu_C - \mu_{M4}}{s_p} J, \quad s_p = \sqrt{\frac{(n_C - 1)s_C^2 + (n_4 - 1)s_4^2}{n_C + n_4 - 2}}, \quad J = 1 - \frac{3}{4(n_C + n_4) - 9}. \quad (2)$$

Bands are negligible (n), small (s), medium (m), large (l), and very large (vl) at 0.2, 0.5, 0.8, and 1.2. Benjamini–Hochberg orders eight $p_{(i)}$ values and retains through the largest index satisfying $p_{(i)} \leq (i/8)q$; * denotes corrected $p < 0.05$.

2 M4 versus eight methods

Table 1: Hedges g ; comparator minus M4.

vs M4	d^*	$T_{personal}$	$T_{mission}$	L	σ_v	N_{stop}
M0	−1.79 vl*	+0.94 l*	−0.74 m*	+0.51 m*	−5.61 vl*	0.00 n
M1	−0.61 m*	+0.67 m*	−0.10 n	−0.22 s	+3.65 vl*	+0.94 l*
M2	−0.63 m*	+0.07 n	−0.04 n	−0.19 n	+4.87 vl*	+2.68 vl*
M3	+0.79 m*	−0.43 s	+0.13 n	+0.15 n	+1.73 vl*	−0.63 m*
B1	+0.12 n	−1.03 l*	+0.52 m	+0.83 l*	+0.98 l*	+2.52 vl*
B2	−2.59 vl*	+1.14 l*	−0.56 m	+0.66 m*	−3.98 vl*	−0.36 s
B3	−0.47 s*	−0.79 m*	−0.19 n	+0.82 l*	−0.84 l*	−0.06 n
B4	+1.76 vl*	−1.35 vl*	+0.11 n	+1.08 l*	+0.93 l*	+0.97 l*

3 Available rank effects

For B2/B3/B4, $\delta = P(X > Y) - P(X < Y)$. The calculations yield large effects: for d^* , $\delta_{B2,B4} = -0.98$, $\delta_{B3,B4} = -0.88$, $\delta_{B2,B3} = -0.81$; for $T_{personal}$, $\delta_{B2,B4} = +1.00$ and $\delta_{B2,B3} = +0.97$; for σ_v , $\delta_{B2,B4} = -1.00$. No Mann–Whitney or Cliff result is invented for comparisons lacking individual data.

4 Conclusion

M4 has shorter paths than all external baselines and is intermediate in safety and smoothness. No baseline dominates M4 across all criteria. The multiplicity correction is the Benjamini–Hochberg procedure.